C04 responsibilities

First integration milestone for the Student Assessment Matrix

CS-352 | DCS/UBIT | Supervisor Humera Tariq | 25 October 2026

Connect A02 student identity with B05 reflection records through C04 assessments and student records. Deliver a tested overview and one reflection activity's six-question detail view. This version adds formative/summative purpose, estimated member effort and clearer reflection headings. Project remains a direct assessment with a separate Bonus rule.

Milestone boundary

A02 owns Student and SectionName. B05 owns Matrix<T> and reflection data. C04 adds nine production classes and one enum (Purpose). Confirm the owners' actual APIs at the recorded commits before relying on them.

Implement one FORMATIVE Reflection activity for three students. Test column growth using test-only Direct objects. Later teams implement examinations, Project, Bonus, verification and marks. The handoff is a tested integration commit.

1 Before implementation

Yes = complete with evidence; No = pending or failed; blank = unchecked.

Check and owner	Evidence link	Yes / No
Affan: Record both PRs and commit IDs. Download those exact code versions.	A02 PR #2 commit: 649211b B05 PR #4 commit: 66ba565	Yes
Affan: Check src/pilot/b05/ and package pilot.b05; in every B05 Java file. Ask Aleeza only if a correction is needed.	GitHub folder and Matrix.java links at that B05 commit: 66ba565 (see p. 10)	Yes
Aown: Check Hafiz's A02 and B05 tester's results: matching commits, java/javac 17 and complete passing output.	Hafiz's test-comment link: not available B05 tester's comment link: PR #4 comment	Yes
Anas Nasir: Update C04_UML.drawio using Sections 4 and 5. Share the draft in C04's Teams thread.	UML file link on Teams: Teams post	Yes
Abdul Rehman: Prepare C04_API_REGISTER.xlsx using Section 5. Match the UML and share it in C04's Teams thread.	API file link on Teams: not available	Yes
All seven: Confirm GitHub access and your two assigned tasks in C04's Teams thread, using your own name.	Link to seven confirmations: see below https://github.com/hum32-cs352/rating-app/issues/3	Yes

For No: report A02 gaps to Anas Arshad on PR #2 and B05 gaps to Aleeza on B05's PR. Only affected work waits. C04 design and independent work may continue.

Working files:

- ✓ Open given C04_UML.drawio at https://app.diagrams.net and refine them to align it with your final code.
- ✓ Edit the shared c04_responsibilities.pdf if needed and submit back with your code.
- ✓ Record agreed changes and their PR discussion links under "Changes made" in README.md.
- ✓ Submit the updated responsibilities guide, editable UML and API register.

Recorded by Affan Rauf (coordinator), Section 1 evidence:
A02 PR #2 commit: 649211bdc5bfc6a815204c403c4530946c2e1e3d
B05 PR #4 commit: 66ba565c4a827ef58e988319c66727d7c4c0bd23
Package check: Yes. Confirmed with Aleeza that B05 uses src/domain/reflection/ (package domain.reflection). Reported on PR #4:
https://github.com/hum32-cs352/rating-app/pull/4#issuecomment-6542944325

2 Students activities and matrix views

Rows are students; columns are assessment activities. A cell holds that student's AssessmentRecord for that activity, or null if missing. S1/S2/S3 are teaching labels; official seat numbers remain the row IDs.

Use these Section C students: S1 Affan — B25110006008; S2 Aown Ali — B25110006026; S3 Abdul Rehman — B25110006004. Names and seats are checked against the team tracker and supplied CS-351 result sheet. Do not import its grades into CS-352.

Example of the growing overview

Student row	A1 Reflection	A2 Midterm	A3 Final Lab Examination
S1 · B25110006008	record or null	record or null	record or null
S2 · B25110006026	record or null	record or null	record or null
S3 · B25110006004	record or null	record or null	record or null

This layout is 3 × 3. C04's implemented pilot is 3 × 1: A1 Reflection. A2/A3 are later modules, represented only by test objects in C04's scaling check. Use FORMATIVE for A1 and SUMMATIVE for A2/A3. These examples are test data.

A1 is a stable activity ID, not array index 1. Store it once and do not renumber after sorting or selection. Pair every ID with getName(): A1 Reflection; A2 Midterm; A3 Final Lab Examination. Dates belong to Schedule.

IDs are unique within one course offering: DCS/UBIT, courseCode="CS-352", semester="2026-Semester-II", SectionName.C. Different offerings may each have A1; keep their matrices separate.

View	Dimensions	What each cell contains
AssessmentMatrix	N × M	AssessmentRecord or null for each activity
AssessmentView	N × K	Same records for K selected activity IDs
ReflectionMatrixView	N × 6	One answer or null for one Reflection activity

One reflection activity expanded into six answers

A1 detail seat	Q1: Study Pattern	Q2: Consultations	Q3: Key Issue	Q4: Reading Habit	Q5: Coding Strength	Q6: Expected Midterm Grade
B25110006008	—	None	—	—	Positive	B
B25110006026	—	Once	Attendance	Videos	Struggling	C
B25110006004	—	—	—	—	—	—

Test data only: two forms received; Abdul Rehman's form is absent in this example. "—" displays a missing answer or form (null). "None" is a selected answer to Q2, not missing data.

The A1 column links to these answers; it does not become six overview columns. Select [A3, A1] for an N × 2 overview. Open A1 detail for N × 6. Build a new snapshot when the roster or activity list grows.

3 Member responsibilities

Java files: src/pilot/c04/assessment/ — package pilot.c04.assessment
Handoff files: handoff/C04/assessment/ — Affan is the sole publisher.

Each *Check.java has public static void run(). Compare actual values and references; throw AssertionError on failure. Only runners print output.

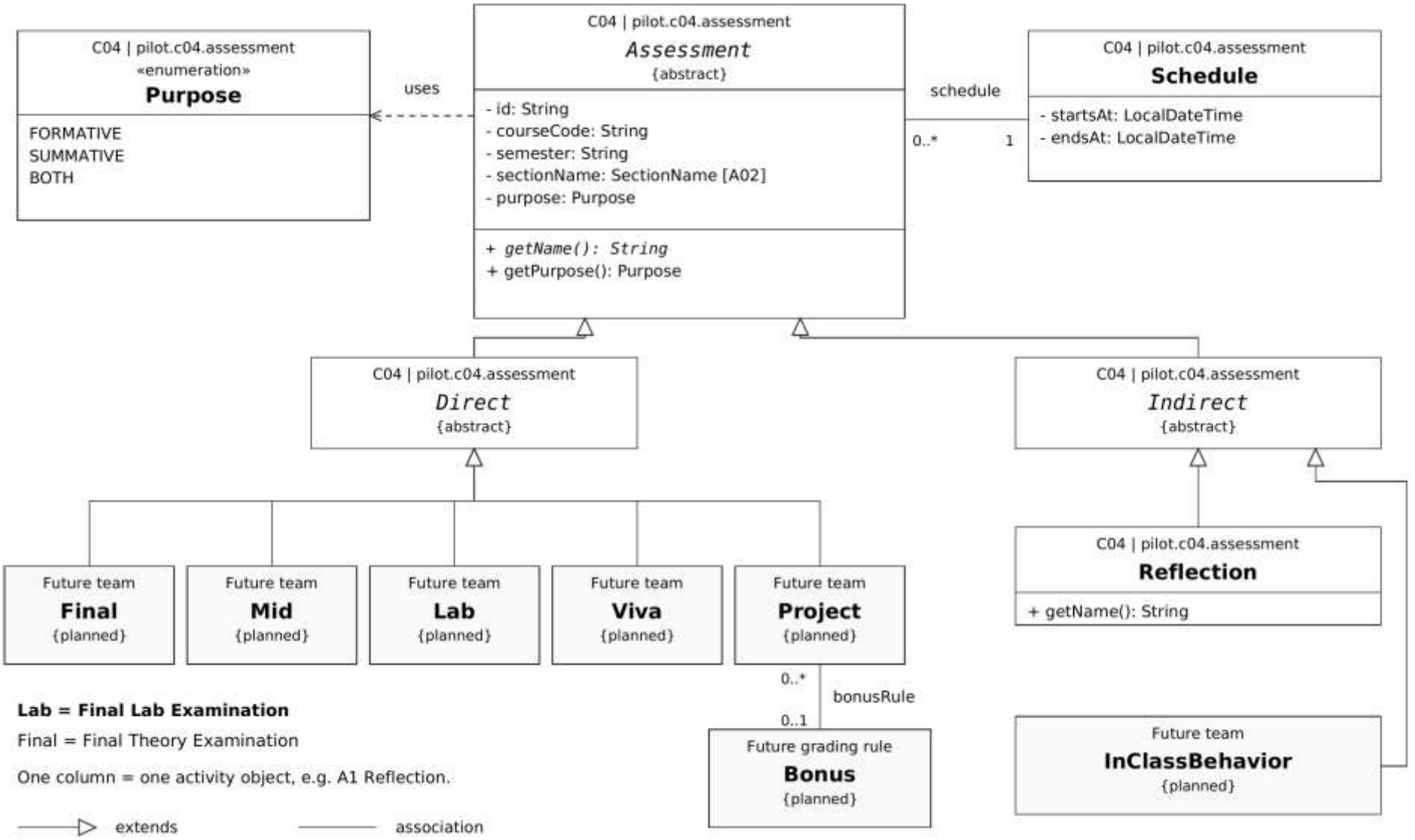
Effort = estimated focused hours for (i) and (ii) together, including learning, tests and handoff; waiting time is excluded. These are planning ranges, not grading targets. Log actual hours in contributions.txt; tell Affan early if more time or support is needed.

Member and estimated effort	Two required tasks	Done Yes / No
Abdul Rehman 6–8 hours	(i) Implement Assessment.java and Schedule.java using Section 5. (ii) Write ScheduleCheck.java for valid, null, equal and reversed times. Maintain C04_API_REGISTER.xlsx from each owner's final declarations.	**Yes**
Asad 4–6 hours	(i) Implement Purpose.java, Direct.java, Indirect.java and Reflection.java. Demo A1 uses FORMATIVE; getName() returns "Reflection". (ii) Write ReflectionCheck.java through Assessment references: check name, ID, context, schedule, all three purposes and rejection of null purpose.	**Yes**
Abeer 4–6 hours	(i) Implement AssessmentRecord.java with private final Student and Assessment references. Reject different sections. (ii) Write RecordCheck.java using ReflectionRecord: check shared references and polymorphic evidence access.	**Yes**
Anas Nasir 6–8 hours	(i) Implement ReflectionRecord.java using A02 Student and B05 StudentPortfolio. Check seat/course/semester; retain optional submittedAt. (ii) Write LinkCheck.java for matching/mismatching records. Maintain C04_UML.drawio to match the code and Section 4.	**Yes**
Moonis Nasir 8–12 hours	(i) Implement AssessmentMatrix.java and AssessmentView.java using B05 Matrix<AssessmentRecord>. Keep the overview array private. (ii) Write MatrixCheck.java for Section 6, including mixed columns, [A3, A1] selection and purpose filters. Use nested test-only classes for future direct records.	**Yes**
Affan Rauf 6–10 hours	(i) Write AssessmentDriver.java for the three-student reflection pilot and its six-question detail. Supply sources.txt, run_assessment.bat and README.md. (ii) Collect actual hours and personal evidence. Publish the module and updated guide/UML/API register; coordinate owner fixes and the final handoff.	**Yes**
Aown 6–10 hours	(i) Write AssessmentChecks.java: main calls all five check classes; print PASS only after each returns successfully. (ii) Independently run Section 6 on JDK 17, record evidence and recommend READY TO MERGE or BLOCKED. Retest the merged integration commit for handoff.	**Yes**

Each member posts a personal C04 PR comment with file link, tested commit, complete output and one explanation. Be ready to change one input and predict its result. Done means both tasks and evidence are complete.

4 UML of assessment categories

Direct/Indirect classify evidence; Purpose describes its use. A hollow triangle points to the superclass. C04 implements its labelled boxes; future boxes are extension points.

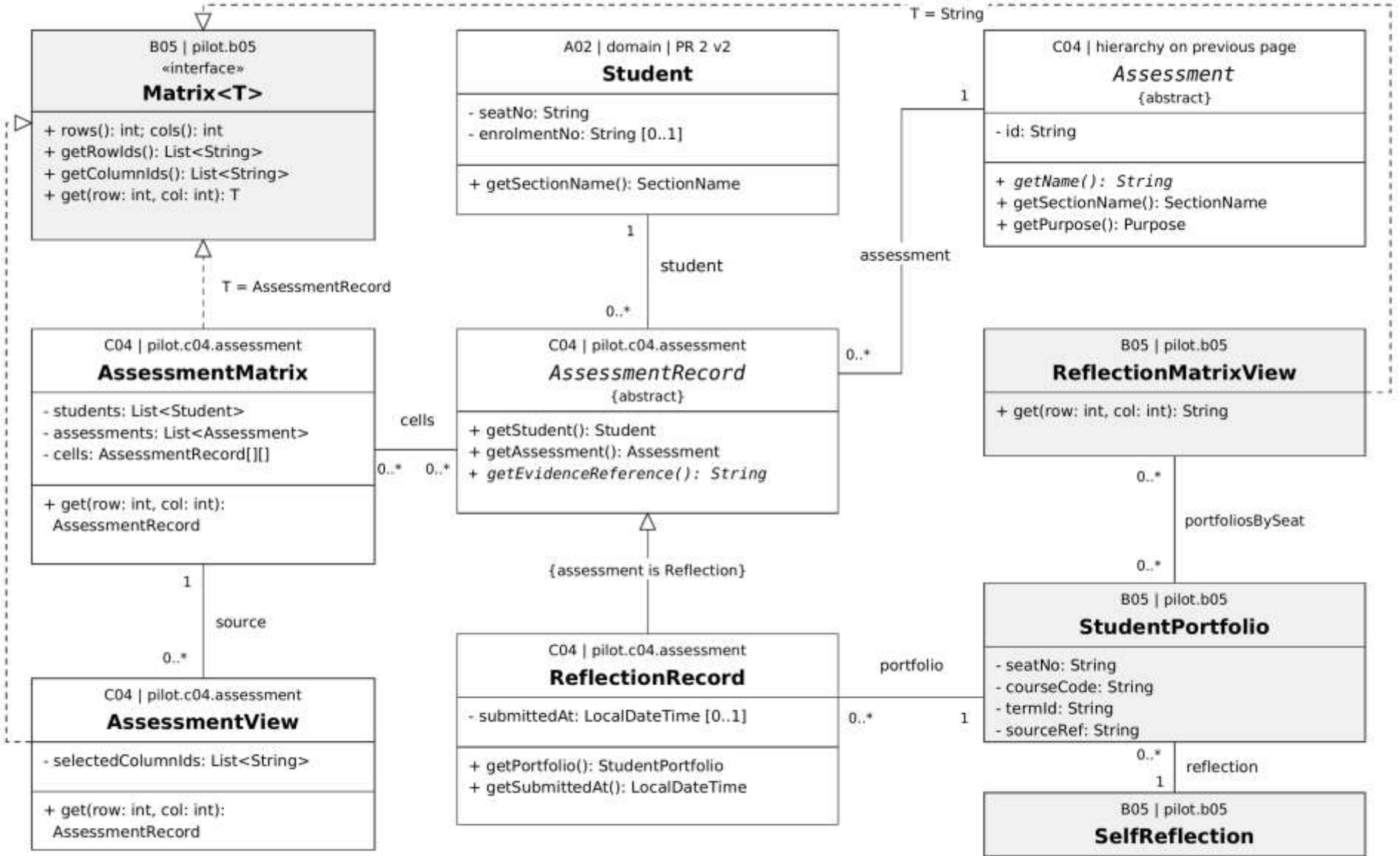


FORMATIVE supports improvement through feedback; SUMMATIVE judges achievement for a grade or final outcome; BOTH serves both purposes. Project checkpoints can be FORMATIVE and the final graded demonstration SUMMATIVE.

Project provides direct evidence; Bonus controls credit. InClassBehavior here means attitude/engagement; evaluated skills are Direct. BOTH keeps one activity and does not double marks.

4 UML of students records and matrices

Same architecture, second view. Plain lines show associations and multiplicities. Dashed hollow triangles implement Matrix. Solid hollow triangles extend a class. Selected members are shown; Section 5 defines the C04 API.



Plain line = association. Student is not a superclass.

One overview per course, semester and section; all matrices reuse B05 Matrix<T>.

1 one; 0..* any number; 0..1 optional. + public; - private. Italic = abstract.

5 Required API and data rules

C04 classes are public; fields private final; getters public. Abstract constructors are protected; concrete constructors public. Concrete classes are final. Assessment requires a non-null Purpose. Use domain.Student, domain.SectionName and the existing pilot.b05 types.

C04 class	Exact constructor and public methods
Purpose enum	public enum Purpose { FORMATIVE, SUMMATIVE, BOTH } Save as Purpose.java in pilot.c04.assessment.
Schedule	Schedule(LocalDateTime startsAt, LocalDateTime endsAt) LocalDateTime getStartsAt(); LocalDateTime getEndsAt() Required java.time values; endsAt strictly after startsAt. Interpret as Karachi local time.
Assessment abstract	Assessment(String id, String courseCode, String semester, SectionName sectionName, Purpose purpose, Schedule schedule) String getId(); abstract String getName(); String getCourseCode(); String getSemester(); SectionName getSectionName(); Purpose getPurpose(); Schedule getSchedule()
Direct and Indirect abstract Reflection	Direct/Indirect extend Assessment; Reflection extends Indirect. All three constructors take the same six parameters as Assessment. Reflection implements getName() returning "Reflection"; other getters are inherited.
AssessmentRecord abstract	AssessmentRecord(Student student, Assessment assessment) Student getStudent(); Assessment getAssessment(); abstract String getEvidenceReference() Required references; matching getSectionName(). Retain supplied objects.
ReflectionRecord	ReflectionRecord(Student student, Reflection assessment, StudentPortfolio portfolio, LocalDateTime submittedAt) extends AssessmentRecord StudentPortfolio getPortfolio(); LocalDateTime getSubmittedAt(); String getEvidenceReference() Required portfolio. Time may be null. Evidence getter delegates to portfolio.getSourceRef().
AssessmentMatrix	AssessmentMatrix(String courseCode, String semester, SectionName sectionName, List<Student> students, List<Assessment> assessments, List<AssessmentRecord> records) implements pilot.b05.Matrix<AssessmentRecord> Five Matrix methods, context getters below, Student getStudent(int row); Assessment getAssessment(int col). Copy ordered lists; keep private AssessmentRecord[][] cells.
AssessmentView	AssessmentView(AssessmentMatrix source, List<String> selectedColumnIds) implements pilot.b05.Matrix<AssessmentRecord> Same public methods as AssessmentMatrix. Copy selection IDs, map to source indexes and retain source; return the original objects.

Five Matrix methods: int rows(); int cols(); List<String> getRowIds(); List<String> getColumnIds(); AssessmentRecord get(int row, int col).

Context getters on both C04 matrices: String getCourseCode(); String getSemester(); SectionName getSectionName(). AssessmentView delegates to its source.

A view preserves every row and requested column order. New selection → new view; changed data → new snapshot. No duplicate Matrix.java or public set/add/delete/update/clone methods.

5 Required API and data rules continued

Reuse these owner contracts

Owner type	C04 uses
A02 Student	Existing constructors; getSeatNo(); getSectionName(). Retain seat-based equals/hashCode. getEnrolmentNo() is additional information and may be null.
B05 Matrix<T>	rows(); cols(); getRowIds(); getColumnIds(); get(row,col). ID lists are unmodifiable. T is AssessmentRecord for C04; String for B05.
B05 StudentPortfolio	Constructor: (String seatNo, String courseCode, String termId, String sourceRef, SelfReflection reflection). Getters: getSeatNo(), getCourseCode(), getTermId(), getSourceRef(), getReflection().
B05 SelfReflection	Constructor: (Map<ReflectionQuestion,String> answers, String remarks). getAnswer(ReflectionQuestion) returns String or null; getRemarks() returns String. Preserve answers.
B05 ReflectionMatrixView	Constructor: (String courseCode, String termId, List<String> seatNos, List<StudentPortfolio> portfolios). Implements the five Matrix<String> methods. Keep B05 storage and validation.

Mandatory rules

1. Link one student to one activity. Compare Student seat to portfolio seat, assessment course to portfolio course, and assessment semester to portfolio termId exactly. Student and assessment sections must match. B05 needs no new section field.

2. B05 portfolios do not contain an assessment ID. The Reflection passed to ReflectionRecord supplies it. The caller selects the correct form; never infer the activity from sourceRef.

3. Build one offering’s overview: copy the roster and activity lists; allocate [students.size()][assessments.size()]. Students must match its section; activities must match course, semester and section. Place records by exact seat plus activity ID; require the same Assessment object as its column.

4. Reject required nulls, blank/padded C04 IDs, null lists/elements, duplicate seats/column IDs/cells, wrong context and records outside the roster/columns with IllegalArgumentException. Reject null source/selection and duplicate/unknown/null selected IDs.

5. Expose no mutable lists or arrays; copy selection IDs. Invalid indexes throw IndexOutOfBoundsException. Missing cells return null. Empty rosters, activities and selections are valid. Dimensions come from the lists, not from the array’s first row.

6. ReflectionRecord submittedAt is the actual submission time or null if unknown. Retain late forms. sourceRef identifies the form or office record; it is not its answers or marks. Receipt does not mean faculty approval. Preserve "None" versus null.

7. Select purpose views using getAssessment(col).getPurpose(). A FORMATIVE filter includes FORMATIVE and BOTH; a SUMMATIVE filter includes SUMMATIVE and BOTH. Pass each matching ID once to AssessmentView. Keep the original column order.

Activity IDs and official seat/enrolment values are opaque strings: do not parse or invent formats. Keep A02/B05 exception contracts. Future record types must be immutable and validate their activity type. A02 Faculty, Section and Office remain outside this slice.

6 Acceptance checklist and Git handoff

Aown runs all checks on JDK 17. Record test ID, expected result, actual result and PASS/FAIL in test-results.txt. Failed or unrun checks block readiness.

ID	Required result	Yes / No
T01	Real A02 v2, B05 and C04 compile/run. Shared starter, A02 runner/new drivers and B05 demo pass. Record dependency commits and java/javac versions.	**Yes**
T02	Schedule accepts valid times; null/equal/reversed endpoints fail. Required null references and blank/padded C04 identifiers fail.	**Yes**
T03	ReflectionRecord retains supplied Student, Reflection and portfolio objects. Wrong seat/course/semester/section fails; evidence getter equals B05 sourceRef.	**Yes**
T04	Through Matrix<AssessmentRecord>, the pilot is 3 × 1: the three official seats, A1 Reflection, two forms and one missing cell as Section 2. Reordering input records changes no cell.	**Yes**
T05	Missing record/answer stays null; the chosen answer "None" stays "None". Original B05 answers are unchanged.	**Yes**
T06	Duplicate seats/IDs/cells and wrong offering context fail. Outside-roster records or unknown activities fail. A different Assessment object from the column fails even when its ID matches.	**Yes**
T07	All indexed methods reject negative/excessive indexes. Select [A3, A1] in the mixed 3 × 3 test: return 3 × 2, retain rows and record references; map getAssessment correctly.	**Yes**
T08	Changes to input/selection lists cannot change either matrix. Returned IDs reject edits. Unknown/duplicate/null selected IDs fail; backing array stays private.	**Yes**
T09	Overview 0 × 0, 3 × 0 and 0 × 3 work. Empty selection gives N × 0. B05 empty detail remains 0 × 6; invalid indexed calls fail.	**Yes**
T10	Extract only A1 ReflectionRecord portfolios; pass all three seats to B05. Detail is 3 × 6: Affan consultations="None", Aown="Once", and all six cells null for the missing form.	**Yes**
T11	Unknown submission time displays "Not recorded"; late forms remain. Received forms show "Received / verification pending". No marks/stars/bonus calculation.	**Yes**
T12	Use nested test-only Direct and record classes for A2 Midterm and A3 Final Lab Examination beside A1. Mix records; leave one Direct column empty. Select Direct activities by type with no matrix code change.	**Yes**
T13	Deliberately fail one check in a disposable local copy. Runner exits nonzero and skips the demo. Restore and pass all checks.	**Yes**
T14	All Purpose values are retained; null fails. In a separate test matrix with A1 FORMATIVE, A2 SUMMATIVE, A3 BOTH, purpose filters select [A1,A3] and [A2,A3]. A3 appears once in each view; records and source matrix are unchanged.	**Yes**

Every member explains a real check and predicts one changed input. Fixed PASS printouts are insufficient. Negative tests catch only the expected exception; any other outcome fails.

Evidence (Aown, 31 Oct 2026, Windows, java/javac 17.0.20.1): test-results.txt. Run on pre-final C04 folder (no git history); A02 commit 649211b / B05 commit 66ba565 per Section 1. Rerun on the final commit pending; link to PR comment: ___

Affan Rauf: all checks above were tested on a local copy; waiting for the A02/B05 merge to test the final folder.

6 Acceptance checklist and Git handoff continued

Publish and verify

1. Affan creates team/C04-assessment from integration. While A02/B05 are pending, test their recorded versions in a separate local copy. Keep their files and unmerged commits out of C04's PR.

2. run_assessment.bat resolves the repository root, cleans only out/C04-assessment, compiles sources with javac --release 17, then runs AssessmentChecks and AssessmentDriver. Stop on failure. List A02 Student/SectionName, six B05 files and seventeen C04 Java files (including Purpose.java) once each.

3. After local checks pass, Affan opens a draft PR into integration with the handoff folder below. A02 corrections go to Anas Arshad on PR #2; B05 corrections to Aleeza on B05's PR. All C04 members post personal evidence on C04's PR.

4. After A02/B05 merge, update C04 from integration. Aown tests a fresh checkout of Affan's final uploaded commit. Post git rev-parse HEAD, java -version, javac -version, commands and complete output on C04's PR. Recheck subsequent code changes.

5. Aown recommends READY TO MERGE or BLOCKED, linking tested commit and findings. Affan marks Ready for review after required checks pass. The supervisor approves and merges.

6. Aown reruns checks on the merged integration commit. Affan announces that passing commit as the baseline for the next teams. Keep the result on C04's PR.

Deliver one handoff folder

`handoff/C04/assessment/`

C04_responsibilities_edited.pdf • C04_UML_updated.drawio • C04_API_REGISTER.xlsx • sources.txt • run_assessment.bat • README.md • test-results.txt • contributions.txt

README records dependencies, commands, offering context, activity IDs/names/purposes and agreed changes with PR links. Complete this guide's checklists with evidence links and submit it. Post final commit/output evidence on the PR; this avoids another code version just to record its ID.

Personal PR comment: Name | Tasks (i)/(ii) | File links | Tested commit | Complete output | Explanation | Done Yes/No.

- ✓ Download **C04_UML.drawio** from my previous message.
- ✓ Open app.diagrams.net.
- ✓ Select **File → Open From → Device**, then choose the downloaded file.

Section 1 evidence links

Added by Affan Rauf (coordinator). Full links for the short labels on page 1.

Row 2: package check (domain.reflection confirmed with Aleeza), PR #4 comment

https://github.com/hum32-cs352/rating-app/pull/4#issuecomment-6542944325

Row 3: Hafiz's A02 test comment

not available

Row 3: B05 tester's comment (PR #4)

https://github.com/hum32-cs352/rating-app/pull/4#issuecomment-6473994732

Row 4: UML draft shared on Teams

https://teams.microsoft.com/l/message/19:2rwcZqXCcMxEDXO3yIOS6vb3MGJNlZ25IG8AUrVAHDk1@thread.tacv2/1793460247869?groupId=62ccd8e9-96ab-44b0-870b-9fe44bb46c13&parentMessageId=1792596982498&tenantId=75df096c-8b72-48e4-9b91-cbf79d87ee3a

Row 5: API register on Teams

not available

Row 6: seven confirmations (Issue #3)

https://github.com/hum32-cs352/rating-app/issues/3

Row 2 note: package name (guide says pilot.b05, real code is domain.reflection)

Issue. The guide (Sections 1 and 5) names the B05 package as pilot.b05, in src/pilot/b05/. The B05 code on PR #4 (commit 66ba565c4a827ef58e988319c66727d7c4c0bd23) is in src/domain/reflection/ and declares package domain.reflection in all six files: Matrix, ReflectionDriver, ReflectionMatrixView, ReflectionQuestion, SelfReflection and StudentPortfolio. A02 (PR #2, commit 649211bdc5bfc6a815204c403c4530946c2e1e3d) uses package domain in src/domain/.

Resolution. Affan checked every B05 file at the recorded commit and confirmed with Aleeza (B05 owner) that domain.reflection is the final package and folder (PR #4 comment, link above). C04 follows the real code and did not edit or move any A02 or B05 file. C04 imports domain.Student, domain.SectionName, domain.reflection.Matrix, domain.reflection.StudentPortfolio, domain.reflection.SelfReflection, domain.reflection.ReflectionQuestion and domain.reflection.ReflectionMatrixView. C04's own package stays pilot.c04.assessment, as the guide requires.

Updated to match. sources.txt lists the B05 files under src/domain/reflection/; README.md records the change under Changes made with the PR #4 link; C04_API_REGISTER.xlsx lists the real packages; C04_UML_updated.drawio labels the B05 boxes domain.reflection. The UML pictures on pages 4 and 5 of this guide are the original images and still show pilot.b05, so the .drawio file is the corrected version. The Section 5 text that says pilot.b05.Matrix refers to domain.reflection.Matrix in the final code.

Result. The Section 1 check was done: the real package and folder were verified at the recorded commit, confirmed by the owner, and C04 compiles and passes its checks against those exact files (tested on a local copy; final folder to be retested after the A02/B05 merge).